# Cover Page

- Problem Statement/Concept Title **AURASIGHT-- Intelligent assistant for visually impaired**
- Theme **Healthcare and Wellbeing**
- Team Name (Registered on portal) **Iotronics**
- Team ID(Registered on portal) **HSC|TN|01897**

- State/UT **TAMIL NADU**

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# Problem – Proposed Solution

Visually impaired people often find it difficult to read everyday text, such as books, signboards, and product labels, without help. Existing refreshable Braille devices can be expensive and may not be affordable for everyone. BRAILIO aims to solve this problem using smart glasses with a camera to capture text, OCR to recognize it, and a portable refreshable Braille display to convert it into touch-readable dots, helping users access information more independently.

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# Technical Approach

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1. **Image Capture**: Smart glasses use a camera to capture printed or digital text.
2. **Text Recognition**: OCR technology extracts text from the captured image.
3. **Braille Conversion:** The recognized text is converted into Braille code.
4. **Wireless Transmission**: The Braille data is sent wirelessly to a portable device.
5. **Refreshable Braille Display**: Small pins rise and lower to form Braille dots.
6. **Touch-Based Reading:** Users read the displayed text through their fingertips, one line at a time.
7. **Independent Access:** The system helps visually impaired people access everyday information with less dependence on others.

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# Impact Potential

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1. **Potential Impact on the Target Audience**
   - Helps visually impaired people read everyday text independently.
   - Reduces the need to depend on others for reading.
   - Improves access to education, public information, and daily activities.
   - Supports greater privacy and confidence.

2. **Benefits of the Solution**
   - Social Benefits - Promotes independence and inclusion.
     Improves access to information and learning.
   - Economic Benefits - Aims to make Braille technology more affordable.
     May reduce long-term dependence on external reading assistance.
   - Educational Benefits - Helps students access books, notes, notices, and learning materials.

3. **Key Stakeholders**
   - Visually impaired students and adults.
   - Blind schools and special education centres.

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# Feasibility and Risk Analysis

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**1.Feasibility**

- Uses existing camera, OCR, and wireless technologies.
- Prototype can be developed and tested.

**2.Challenges & Risks**

- High Braille display cost.
- OCR errors and hardware issues.

**3.Solutions**

- Use low-cost components.
- Improve OCR through testing.
- Test and optimize the device in stages.

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# Team

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## Team Members & Roles

- Haritha R (ECE) – Team Lead: Team coordination and project planning.
- Anudeeksha B (IoT): IoT integration and wireless communication.
- Mahalakshmi S (ECE): Hardware design and prototype development.

## Challenges

- Integrating hardware and software.
- Developing an affordable Braille display.
- Coordinating tasks across different specializations.

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# References


1. **WHO – Blindness and Vision Impairment**
https://www.who.int/news-room/fact-sheets/detail/blindness-and-vision-impairment
2. **WHO – Assistive Technology**
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3. **Google Scholar – OCR and Assistive Technology for Visually Impaired People**
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4. **Google Scholar – Refreshable Braille Displays**
https://scholar.google.com/scholar?q=refreshable+Braille+display+research